Assignment 2: Data Processing with Databricks Spark

Big Data Engineering Spring 2025 Session

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Section 1: Aims and Objectives

1.1 Introduction

Taxis are a vital part of New York City's transport network, connecting millions of people to work, leisure, and everyday life. Since 1971, the New York City Taxi and Limousine Commission (TLC) has overseen the city's taxi services, regulating both the famous yellow cabs and the newer green cabs that serve areas outside Manhattan. To improve transparency and encourage innovation, the TLC has released millions of trip records spanning more than a decade. These records capture details such as pick-up and drop-off times and locations, trip distances, fares, passenger counts, and payment methods.

This assignment aims to use SQL/Spark in Databricks to extract, clean, and analyse this dataset, and then apply machine learning to predict trip outcomes.
This process involves several steps:

1. First, we will carry out data ingestion and cleaning to remove trips with unrealistic values, such as negative distances or impossible speeds, and do general data cleaning on the dataset.
2. Next, we answer a series of business-oriented questions using SQL, including identifying when and where most trips occur, how much passengers pay, and which routes generate the most revenue. This helps analyse data and draw meaningful insights for the business.
3. Finally, we build predictive models to estimate the total fare of a trip, comparing different approaches to see which provides the most accurate and efficient results.

Each section of this report will provide a clear overview of the steps taken, present insights and business recommendations, and reflect on the challenges and lessons learned from the dataset.

1.2 Introduction to the Dataset

The dataset at the centre of this project is the New York City Taxi and Limousine Commission (TLC) Trip Record Data, one of the most detailed public transport datasets in the world. It contains millions of records of trips made by the city's iconic yellow cabs and the green cabs (introduced in 2013 to improve access in outer boroughs).

The Dataset captures a variety of information for each trip. Every record reflects a single passenger journey, with details including:

1. Pick-up and drop-off times and locations
2. Trip distance and duration
3. Fares, tips, surcharges, and the 2025 congestion fee
4. Passenger count and payment type, etc.

In addition to taxis, recent years also included For-Hire Vehicle (FHV) and High Volume FHV (HVFHV) data, capturing services like Uber and Lyft. In 2025, a new CBD congestion fee was added to reflect policy changes in managing traffic in Manhattan's busiest zones. By analysing patterns across time, location, and fare structures, the data offers insights into how New Yorkers travel, how drivers earn, and how regulations shape the city's transport ecosystem.

1.3 Project Aims

1. Upload and organise the raw CSV and JSON files into Snowflake using a Data Lakehouse approach.
2. Clean and transform the dataset by removing duplicates, handling missing values, and ensuring consistency across files.
3. Analyse video performance across countries and categories using engagement metrics such as views, likes, and comments.
4. To address a business-driven question by identifying the most strategic content category for launching a new YouTube channel and assessing its global applicability.

Section 2: Approach Overview:

1. **Data Exploration:** I started by familiarising myself with the raw dataset, examining its structure, key features (trip times, distances, fares, passenger counts), and potential issues to understand the opportunities for analysis.
2. **Data Preparation and Cleaning:** Using Spark and SQL, I removed unrealistic trips (e.g., negative speeds, trips ending before they started, or abnormal distances), and then integrated yellow and green taxi datasets with location data to create a clean, enriched dataset ready for analysis.
3. **Analysis and Business Insights:** I used SQL to answer key business questions, focusing on trip volumes, passenger patterns, fare structures, and revenue distribution, summarised across time, location, and taxi type to generate actionable insights.
4. **Machine Learning and Evaluation:** I built and compared two predictive models for trip fares, evaluated their performance using RMSE against a baseline, and reflected on challenges like feature selection and handling large-scale data, documenting solutions and learnings throughout the process.

Section 3: Tasks: Part 1: Data Ingestion and Exploration

1.1 Loading the Data:

1. **Loading Data:** Imported the yellow and green taxi datasets in Parquet format into Databricks and manually loaded the location referential CSV.
2. **Configurations**: Configured a 32 GB serverless cluster in Databricks Free Edition to handle the large datasets efficiently. Set up the cluster connections and added additional dependencies, such as the Scikit-Learn library, ensuring Databricks could process, clean, and perform key analysis and modelling on the data.

Green Columns
VendorID
lpep_pickup_datetime
lpep_dropoff_datetime
store_and_fwd_flag
RatecodeID
PULocationID
DOLocationID
passenger_count
trip_distance
fare_amount
extra
mta_tax
tip_amount
tolls_amount
ehail_fee
improvement_surcharge
total_amount
payment_type
trip_type
congestion_surcharge

Yellow Columns
VendorID
tpep_pickup_datetime
tpep_dropoff_datetime
passenger_count
trip_distance
RatecodeID
store_and_fwd_flag
PULocationID
DOLocationID
payment_type
fare_amount
extra
mta_tax
tip_amount
tolls_amount
improvement_surcharge
total_amount
congestion_surcharge
airport_fee

1.2 Initial Dataset Count:

1. Green taxi dataset: 83,484,688 records
2. Yellow taxi dataset: 907,982,776 records
3. Combined total: 991,467,464 trip records

1.3 Feature Understanding:

1. **Trip and Passenger Details:** Includes pickup and drop-off times, trip distance, passenger count, rate type, and trip classification (green taxis).

2. **Payment and Location Information:** Covers fares, tips, tolls, surcharges (including congestion and airport fees), payment type, and pickup/drop-off location IDs, providing a complete view of each trip.

1.4 Data Cleaning:

The data cleaning was performed in a step-by-step loop format, applying each cleaning rule separately to both the yellow and green taxi datasets. After every cleaning step, the new data was stored in a new cleaned DataFrame for each dataset. This ensured that each subsequent cleaning step was applied to an already validated dataset, maintaining data integrity throughout the process.

1. Trips with Invalid Time: Removed trips that finished before the starting time and trips where pickup or drop-off datetimes were outside the expected range.

2. Trips with Unrealistic Speed or Distance: Filtered out trips with negative or extremely high speeds (above NYC limits). Removed trips with very short or very long distances and durations that were not plausible.

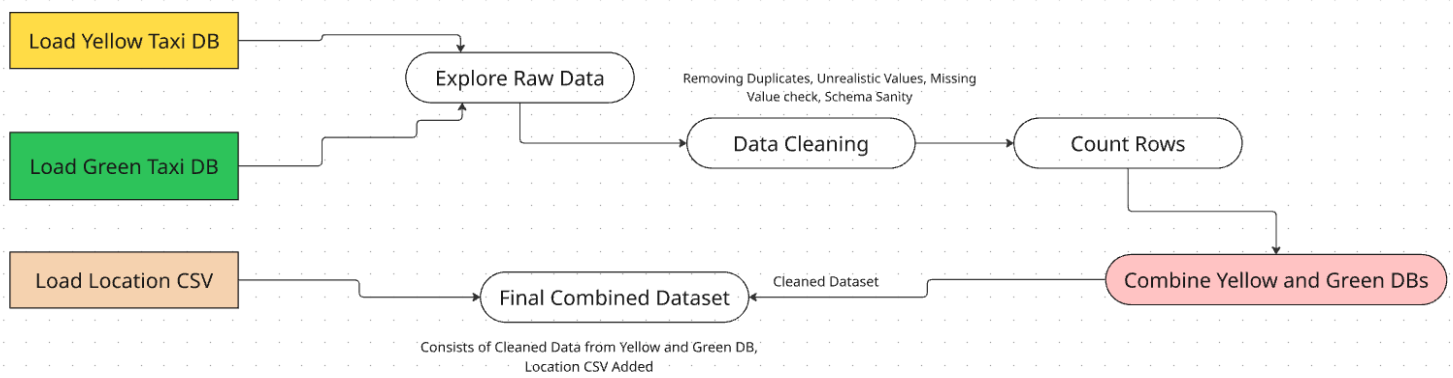
3. Passenger Count Validation: Excluded trips with unrealistic passenger counts, such as 0 passengers.

4. Duplicate Records: Identified and removed duplicate trip records to avoid redundancy in the dataset.

5. Handling Missing Values: Left missing values as they were because they exceeded 10% of the dataset. The decision to impute missing data will be made later during feature engineering if the feature is deemed important.

1.5 Dataset Count After Cleaning:

1. Green taxi dataset: 81,000,695 records
2. Yellow taxi dataset: 895,067,506 records



1.6 Merging and Standardizing the Dataset:

1. **Standardizing and Merging:** The yellow and green taxi datasets had different column names and structures, so I standardized them by creating temporary views (green_std and yellow_std) with consistent column names, renaming pickup and drop-off timestamps, and adding missing columns as NULL where needed (e.g., airport_fee for green taxis, ehail_fee and trip_type for yellow taxis).
2. **Taxi Type Identification:** Created a new column, taxi_color, to indicate whether each record came from a green or yellow taxi, enabling analysis by taxi type.

3. **Joining Location Data:** Combined dataset by joining with the location referential CSV, adding borough, zone, and service zone information for both pickup and drop-off locations, resulting in a final cleaned and enriched table ready for analysis.

1.7 Final Dataset:

1. Final taxi dataset (taxi_with_locations_final): 960056472 records

	VendorID	pickup_datetime	dropoff_datetime	store_and_fwd_flag	RatecodeID	PULocationID	DOLocationID	passenger_count	trip_distanc
1	2	2009-01-01T01:21:32.000+00:...	2009-01-01T01:33:09.000+00:...	N	1	223	138	1	
2	2	2010-09-23T00:34:02.000+00:...	2010-09-23T00:50:22.000+00:...	N	1	112	25	1	
3	2	2010-09-23T00:56:57.000+00:...	2010-09-23T01:03:03.000+00:...	N	1	94	264	1	
4	2	2010-09-23T01:11:49.000+00:...	2010-09-23T01:20:47.000+00:...	N	1	41	42	1	
5	2	2010-09-23T01:54:24.000+00:...	2010-09-23T02:19:14.000+00:...	N	1	95	191	1	
6	2	2010-09-23T05:23:27.000+00:...	2010-09-23T05:28:34.000+00:...	N	1	82	173	1	
7	2	2010-09-23T14:40:26.000+00:...	2010-09-23T15:19:36.000+00:...	N	1	191	264	1	
8	2	2010-09-23T23:58:07.000+00:...	2010-09-24T00:01:07.000+00:...	N	1	74	74	1	
9	2	2014-01-01T00:00:00.000+00:...	2014-01-01T02:37:20.000+00:...	N	1	264	226	1	
10	2	2014-01-01T00:00:16.000+00:...	2014-01-01T00:08:32.000+00:...	N	1	226	226	1	
11	2	2014-01-01T00:07:06.000+00:...	2014-01-01T00:15:23.000+00:...	N	1	129	129	5	
12	1	2014-01-01T00:08:09.000+00:...	2014-01-01T00:26:48.000+00:...	N	1	75	244	1	
13	2	2014-01-01T00:12:07.000+00:...	2014-01-01T00:18:48.000+00:...	N	1	248	185	1	
14	2	2014-01-01T00:12:08.000+00:...	2014-01-01T00:24:34.000+00:...	N	1	136	75	1	
15	2	2014-01-01T00:12:27.000+00:...	2014-01-01T00:21:01.000+00:...	N	1	260	226	1	
16	1	2014-01-01T00:12:31.000+00:...	2014-01-01T00:45:35.000+00:...	N	1	75	216	1	
17	2	2014-01-01T00:14:17.000+00:...	2014-01-01T00:16:24.000+00:...	N	1	17	17	1	
18	2	2014-01-01T00:15:46.000+00:...	2014-01-01T00:40:55.000+00:...	N	1	80	68	1	
19	2	2014-01-01T00:15:50.000+00:...	2014-01-01T00:20:59.000+00:...	N	1	61	83	2	
20	2	2014-01-01T00:16:04.000+00:...	2014-01-01T00:20:57.000+00:...	N	1	7	223	1	

Section 4: Tasks: Part 2: Business Questions & Analysis

4.1 Question 1:

Task Performed:

1. Aggregated Monthly Data: Calculated key monthly metrics from the taxi_with_locations_final table, including the total number of trips, average passengers per trip, average amount paid per trip, and average amount paid per passenger.
2. Most Trips by Day: Determined the day of the week (Monday, Tuesday, etc.) with the highest number of trips for each month.
3. Most Trips by Hour: Determined the hour of the day (0–23) when the most trips occurred for each month.
4. Final Output: Joined these results into a single table, providing a monthly overview of trip volume, peak day, peak hour, and fare statistics for analysis.

	year_month	total_trips	most_trips_day	most_trips_hour	avg_passengers	avg_amount_per_trip	avg_amount_per_passenger
1	2009-01	1226	Thursday	0	1.6	18.29	11.45
2	2010-09	204	Thursday	1	1.05	17.64	16.82
3	2011-01	2	Monday	23	1	12.8	12.8
4	2011-02	1	Tuesday	0	1	15.8	15.8
5	2012-09	3	Thursday	19	1	9.79	9.79
6	2014-01	14427669	Friday	19	1.69	14.37	8.48
7	2014-02	13915170	Saturday	19	1.68	14.5	8.62
8	2014-03	16543708	Saturday	19	1.68	14.66	8.74
9	2014-04	15767558	Wednesday	19	1.68	14.95	8.9
10	2014-05	16030454	Friday	19	1.68	15.52	9.24
11	2014-06	14992704	Sunday	19	1.68	15.53	9.24
12	2014-07	14228682	Thursday	19	1.68	15.2	9.03
13	2014-08	13709937	Friday	19	1.68	16.12	9.59
14	2014-09	14564891	Tuesday	19	1.66	15.57	9.38
15	2014-10	15626709	Friday	19	1.66	15.57	9.38
16	2014-11	14674398	Saturday	19	1.66	15.37	9.25
17	2014-12	14569667	Wednesday	19	1.66	15.54	9.34
18	2015-01	14076711	Saturday	19	1.65	15.04	9.09
19	2015-02	13847180	Saturday	19	1.64	15.23	9.29
20	2015-03	14874336	Sunday	19	1.64	15.66	9.54

Insights Drawn:

1. Peak Travel Times: Most trips occur around 19:00 hours, with weekends (Friday–Sunday) and specific weekdays showing higher volumes.
2. Passenger and Fare Patterns: Average passengers per trip remain low (~1.6), while average trip amounts are fairly consistent (~$14–$16 per trip).

3. Monthly and Seasonal Trends: Trip volumes fluctuate by month, with peaks in summer and holiday periods.

Business Recommendations

1. Optimize Driver Shifts: Schedule more drivers during peak hours (evenings) and on high-demand days to reduce wait times and improve service.
2. Encourage Multi-Passenger Trips: Introduce promotions or incentives to increase passenger count per trip and maximize revenue.
3. Plan Fleet Allocation Strategically: Use monthly and seasonal trends to manage fleet distribution, implement surge pricing, and target marketing efforts during high-demand periods.

4.2 Question 2:

Tasks Performed:

1. Calculated trip duration, distance, and speed statistics (average, median, minimum, and maximum) for both yellow and green taxis.
2. Applied filters to remove invalid trips with missing timestamps or negative durations to ensure accurate results.
3. Grouped the data by taxi_color to allow comparison of key trip metrics between yellow and green taxis.

	taxi_color	avg_duration_min	median_duration_min	min_duration_min	max_duration_min	avg_distance_km	median_distance_km	min_distance_km
1	green	13.55	10.50	1.00	359.98	2.96	1.93	0.02
2	yellow	14.38	11.25	1.00	360.00	3.04	1.7	0.02

Insights Drawn:

1. Trip Duration: Yellow taxis have slightly longer average trips (14.38 min) compared to green taxis (13.55 min), with similar median durations.
2. Trip Distance: Yellow taxis also cover slightly longer distances on average (3.04 km vs 2.96 km).
3. Speed Patterns: Green taxis tend to travel at slightly higher average and median speeds (12.68 km/h vs 11.74 km/h), but maximum speeds for both taxi types reach 70 km/h.

Business Recommendations

1. Fleet and Route Planning: Allocate taxis based on typical trip lengths and speeds; yellow taxis may be better suited for slightly longer trips, while green taxis can focus on quicker, shorter routes.

2. Driver Training and Safety: Since maximum speeds reach 70 km/h, enforce speed regulations and provide training to ensure safe driving, particularly in high-traffic areas.

3. Operational Efficiency: Use insights on average duration and distance to optimize scheduling and reduce idle times, especially during peak hours.

4.3 Question 3:

Tasks Performed:

1. Aggregated Trip Data: Calculated the total number of trips, average distance, average fare per trip, and total fare collected for each combination of taxi colour, pickup and drop-off boroughs, month, day of the week, and hour of the day.
2. Filtered Data: Removed records with missing pickup or drop-off boroughs and considered only valid taxi colours (yellow and green) to ensure accurate aggregation.
3. Grouped and Ordered: Grouped the results by taxi colour, pickup and drop-off boroughs, month, day, and hour, then ordered the output to make it easy to analyze patterns across location and time.

	taxi_color	pickup_borough	dropoff_borough	year_month	day_of_week	hour_of_day	total_trips	avg_distance_km	avg_amount_per_trip
1	green	Bronx	Bronx	2009-01	Thursday	0	4	0.5	2.53
2	green	Bronx	Bronx	2009-01	Thursday	3	1	0.51	5.8
3	green	Bronx	Bronx	2009-01	Thursday	9	1	0.73	6.3
4	green	Bronx	Bronx	2009-01	Thursday	10	1	1.43	9.8
5	green	Bronx	Bronx	2009-01	Thursday	15	1	4.42	18.3
6	green	Bronx	Bronx	2009-01	Thursday	16	1	6.05	26.8
7	green	Bronx	Bronx	2009-01	Thursday	17	1	3.31	19.8
8	green	Bronx	Bronx	2009-01	Thursday	18	1	3.04	15.8
9	green	Bronx	Bronx	2009-01	Thursday	23	1	1.56	10.8
10	green	Bronx	Bronx	2010-09	Thursday	0	5	2.28	12.26
11	green	Bronx	Bronx	2010-09	Thursday	1	9	3.88	16.27
12	green	Bronx	Bronx	2010-09	Thursday	2	3	3.18	16.19
13	green	Bronx	Bronx	2010-09	Thursday	3	2	1	8.55
14	green	Bronx	Bronx	2010-09	Thursday	9	3	2.22	15.25
15	green	Bronx	Bronx	2010-09	Thursday	10	1	21.71	89.8
16	green	Bronx	Bronx	2010-09	Thursday	12	1	4.25	18.8
17	green	Bronx	Bronx	2010-09	Thursday	14	1	3.09	45.39
18	green	Bronx	Bronx	2010-09	Thursday	15	1	1.1	13.3
19	green	Bronx	Bronx	2010-09	Thursday	16	2	8.94	35.8
20	green	Bronx	Bronx	2010-09	Thursday	18	1	2.5	10.3

Insights Drawn:

1. High Variability in Trips and Fares: The number of trips and total fares vary significantly by hour, even for the same borough pair. Some hours have very few trips but high fares (e.g., long-distance trips), while others have multiple short trips with lower fares.
2. Peak Hours Identification: For the Bronx → Bronx route, trips occur throughout the day but clustering is visible around late morning to late afternoon (e.g., 10:00–18:00), indicating higher demand during these hours.

3. Distance vs Fare Patterns: Longer trips generally result in higher total fares, but some short trips also have higher fares, likely due to surcharges, congestion fees, or tips.

Business Recommendations

1. Consider implementing time-of-day or distance-based dynamic pricing, especially during peak hours, to maximize revenue.
2. Assign more taxis to high-demand hours and borough pairs to reduce wait times and increase trip completion efficiency.
3. Identify borough pairs with low trip frequency but high fares as potential targets for promotional offers to attract more riders.

4.4 Question 4:

Tasks Performed:

1. Aggregated Revenue by Borough Pairs: Calculated the total revenue generated in 2024 for each pickup dropoff borough pair.
2. Ranked the Pairs: Used SQL RANK() to identify the top 10 borough pairs based on total revenue.
3. Computed Revenue Share: Calculated each top pair's contribution as a percentage of the total 2024 revenue.

	pickup_borough	dropoff_borough	total_revenue	revenue_share_pct
1	Manhattan	Manhattan	629438925.33	61.3
2	Queens	Manhattan	168721993.3	16.43
3	Manhattan	Queens	68036452.59	6.63
4	Queens	Brooklyn	36782607.87	3.58
5	Manhattan	Brooklyn	31244612.75	3.04
6	Queens	Queens	29796861.75	2.9
7	Queens	Unknown	14293710.79	1.39
8	Manhattan	EWR	11720875.78	1.14
9	Manhattan	Unknown	6700299.92	0.65
10	Queens	Bronx	5426693.4	0.53

Insights Drawn:

1. Manhattan-Dominant Revenue: The Manhattan → Manhattan route alone contributes 61.3% of total revenue, showing extremely high demand within central Manhattan.
2. High-Value Cross-Borough Trips: Routes like Queens, Manhattan (16.43%), and Manhattan, Queens (6.63%) are the next largest contributors, highlighting key commuter corridors.
3. Smaller Contributions from Other Boroughs: Brooklyn, Bronx, and EWR trips collectively account for a smaller share (<10%), indicating lower revenue density outside core Manhattan routes.

Business Recommendations

1. Assign more taxis to high-demand Manhattan and Queens, and Manhattan routes to maximize revenue efficiency.

2. Consider pricing incentives or loyalty programs for less-traveled borough routes to improve utilization.

3. Focus on improving service and coverage in high-revenue corridors while monitoring demand trends in secondary routes for potential growth.

4.5 Question 5:

Tasks Performed:

1. Calculated the percentage of trips where drivers received tips by counting all trips with tip_amount > 0 and dividing by the total number of trips.
2. Applied a filter to exclude trips with NULL tip_amount to ensure accuracy.

	.00 pct_trips_with_tips
1	63.04

Insights Drawn:

1. About 63% of all trips included a tip, indicating that tipping is a common practice among passengers in NYC taxis. This suggests that drivers earn a significant portion of income from tips, not just the base fare.

Business Recommendations

1. Taxi companies could consider incentive programs to encourage tips, such as loyalty rewards or service quality recognition.
2. Drivers can focus on providing excellent service and targeting routes with higher tipping likelihood to maximize earnings.

4.6 Question 6:

Tasks Performed:

1. Calculated the percentage of trips where the tip amount was at least $15, considering only trips where drivers received any tip.
2. Used a CASE WHEN statement in SQL to count qualifying trips and divided by the total number of trips with tips to get the percentage.

Table

	.00 pct_trips_tip_15_or_more
1	0.82

Insights Drawn:

1. Only 0.82% of all trips resulted in tips of $15 or more, showing that very high tips are extremely rare in the dataset.
2. The vast majority of tips are smaller amounts, indicating most passengers tip modestly.

Business Recommendations

1. Focus driver training and service quality improvements on long or premium trips, which are more likely to generate larger tips.

4.7 Question 7:

Tasks Performed:

1. Classified each trip into duration bins (Under 5 mins, 5–10 mins, 10–20 mins, 20–30 mins, 30–60 mins, 60+ mins) based on trip duration in minutes.

	duration_bin	avg_speed_kmh	avg_distance_per_dollar	num_trips
1	Under 5 Mins	12.27	0.1	133881726
2	5-10 Mins	10.69	0.13	284499954
3	10-20 Mins	11.13	0.16	339066674
4	20-30 Mins	13.27	0.19	121951088
5	30-60 Mins	16.05	0.23	70803593
6	60+ Mins	14.21	0.4	8163619

2. Calculated average speed and average distance per dollar for each bin to evaluate efficiency and cost-effectiveness.
3. Filtered out trips with missing timestamps, negative durations, zero distance, or zero total amount to ensure accurate calculations.

Insights Drawn:

1. Under 10 mins trips (Under 5 & 5–10 mins) have low distance per dollar (0.1–0.13 km/$), showing short rides are less cost-efficient.
2. Medium trips (10–30 mins) show a balanced speed and better distance per dollar (0.16–0.19 km/$), indicating efficient trips for passengers and drivers.
3. Long trips (30–60 mins and 60+ mins) have the highest distance per dollar (0.23–0.4 km/$), but fewer in number, reflecting long-distance, high-value rides.

Business Recommendations

1. Introduce pricing incentives or bundled fares for short trips to improve efficiency and profitability.
2. Promote medium-duration trips via targeted offers or loyalty programs, as they balance speed, revenue, and customer value.

4.8 Question 8:

Tasks Performed:

1. Classified all trips into duration bins (Under 5 mins, 5–10 mins, 10–20 mins, 20–30 mins, 30–60 mins, 60+ mins).
2. Computed total revenue, average revenue per trip, tip frequency, average tip, and tip percentage for each bin to generate more insights.

Table +

	duration_bin	num_trips	total_revenue	avg_revenue_per_trip	pct_trips_with_tip	avg_tip	avg_tip_pct
1	10-20 Mins	337202495	5490281199.1	16.28	65.55	1.8	0.1
2	30-60 Mins	70552667	3358166318.48	47.6	66.09	5.31	0.1
3	20-30 Mins	121237855	3343657325.38	27.58	66.58	3.12	0.1
4	5-10 Mins	288509003	2950888903.16	10.23	61.83	1.12	0.1
5	Under 5 Mins	134389883	974038833.18	7.25	54.84	0.77	0.09
6	60+ Mins	8164569	583690002.5	71.49	58.48	7.26	0.09

6 rows | 2m 50s runtime

Insights Drawn:

1. 10–20 mins trips:Highest total revenue ($5.49B, 337M trips) driven by very high trip volume, though average revenue per trip is modest ($16.28).
2. 60+ mins trips: Highest average revenue per trip ($71.49), but very low volume (~8M trips, <2% of total).
3. 20–30 mins & 30–60 mins trips: Best balance with high avg revenue per trip ($27.58 and $47.6), strong trip volumes (121M and 70M), and stable tip rates (~66%).

Business Recommendations

1. Primary focus: Target 30–60-minute trips; these maximize income per trip with strong volume, making them highly profitable.
2. Secondary focus: Accept 10–20-minute trips, though less per trip, their huge frequency ensures a steady income flow.

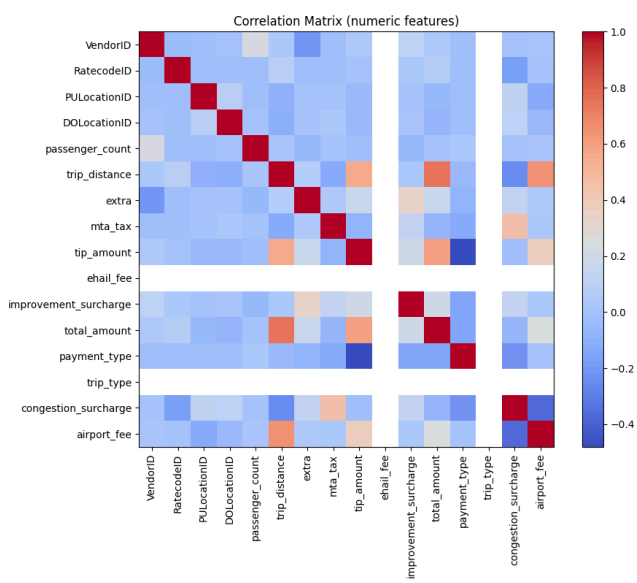
Section 5: Tasks: Part 3: Machine Learning

Tasks:

The goal of this task is to predict the total amount for individual taxi trips using machine learning models. We begin by establishing a baseline, which calculates the average total amount per combination of taxi color, pickup, and dropoff boroughs, month, day of week, and hour. All data except October, November, and December 2024 will be used for training and validation, while these three months will serve as the test set to evaluate predictive performance. Two different algorithms will be applied: Linear Regression with one-hot encoded features and a Random Forest model with label-encoded features. To avoid data leakage, fare_amount and tolls_amount are excluded from the features. Models will be evaluated using RMSE along with considerations of processing time and complexity, and the best-performing model will be used to predict the total amount of trips for October, November, and December 2024, comparing its performance to the baseline to assess improvement.

5.1. Sampling the Dataset:

Since the full taxi dataset is very large, we first created a 1% random sample to make the modeling process faster and to ensure that queries and computations run efficiently in memory. We experimented with larger samples (5% and 10%), but these attempts caused crashes due to memory limits. By limiting to 1% (and optionally capping at 500,000 rows), we can safely convert the Spark DataFrame to a pandas DataFrame for local processing, enabling quick experimentation and model development without overwhelming the system.



5.2. Correlation Analysis:

The correlation analysis shows that trip_distance is the strongest driver of fare amounts, followed by tip_amount, both aligning with intuitive fare dynamics. Moderate positive effects from airport_fee, improvement_surcharge, and extra highlight the role of flat or regulated charges in fare calculation. In contrast, features like passenger_count, location IDs, and congestion_surcharge contribute little explanatory power. A weak negative link with payment_type suggests subtle differences between cash and card trips. Finally, variables like ehail_fee and trip_type appear irrelevant due to lack of variability or missing data.

5.1. ANOVA Test on Categorical Variables:

The ANOVA F-test and Mutual Information results highlight that payment_type has the strongest relationship with total fare, followed by pickup/dropoff zones and location IDs, reflecting the influence of trip geography and payment method on fare variability. VendorID and RatecodeID also show moderate associations, likely tied to company-specific pricing or fare structures. In contrast, variables like store_and_fwd_flag, trip_type, and taxi_color contribute minimally, suggesting they add little explanatory power to predicting total_amount. Overall, categorical effects are driven mainly by payment method and trip location characteristics.

5.2. Feature Selection:

In the feature selection process, I first dropped columns with more than 10% missing values, such as airport_fee and ehail_fee, to avoid introducing bias or noise. Next, I removed columns that were either irrelevant or had very low correlation with the target variable total_amount, including identifiers (VendorID, PULocationID, DOLocationID, RatecodeID) and fare components like fare_amount, tolls_amount, extra, mta_tax, and surcharges (congestion_surcharge, improvement_surcharge). This reduced the dimensionality and retained only features likely to contribute meaningful predictive information, making the dataset cleaner and more suitable for modeling.

5.3. Feature Engineering:

In this step, I extracted time-based features hour, day_of_week, month, and year from the pickup_datetime column to capture temporal patterns in trip pricing. These features help the model account for daily, weekly, and seasonal variations in total_amount. They are then used in encoding for linear and tree-based models.

5.4. One-Hot Encoding:

In this step, we prepared the categorical features for two types of models. For linear models, we applied One-Hot Encoding by creating a binary column for each category in the selected features, generating df_linear_encoded that preserves numeric features while converting all categorical variables into a suitable format for linear regression. For tree-based models, we used Label Encoding, converting each category into an integer code within df_tree, which works naturally with decision trees and ensemble methods. This dual approach ensures that both model types can effectively handle categorical data while maintaining the integrity of the original dataset.

5.5. Outlier Detection:

In this step, we are performing outlier detection on key numeric features. First, we use boxplots to visually inspect the distribution of passenger_count, trip_distance, tip_amount, and total_amount, quickly highlighting extreme values. Then, we calculate the interquartile range (IQR) for each column and determine the lower and upper bounds (1.5×IQR below Q1 or above Q3) to quantify potential outliers numerically. This helps identify unusually high or low values

that might skew our model training. From the IQR-based analysis, we see a substantial number of potential outliers in several key features: passenger_count has 46,748, trip_distance 54,611, tip_amount 25,516, and total_amount 45,546 values flagged as extreme. This indicates that a significant portion of the dataset contains unusually high or low values, which could distort model training and predictions. Recognizing these outliers is important for deciding whether to cap, transform, or remove them before building our regression models.

5.6. Outlier Detection:

This code is capping extreme values (outliers) in the numeric columns passenger_count, trip_distance, and tip_amount to reduce their impact on modeling. For each column, values below the 1st percentile are set to the 1st percentile and values above the 99th percentile are set to the 99th percentile, keeping all data within a reasonable range. After clipping, df[numeric_cols].describe() is used to quickly check the summary statistics, ensuring the data is now more robust and less influenced by extreme outliers.

5.7. Scaling:

Here, we applied feature scaling to standardize the numeric columns passenger_count, trip_distance, tip_amount, and total_amount using StandardScaler, which transforms them to have a mean of 0 and a standard deviation of 1. We used a ColumnTransformer to apply scaling only to numeric columns while leaving all other columns (categorical or already encoded features) unchanged (passthrough). The resulting df_scaled contains a mix of scaled numeric features and untouched categorical features, ready for machine learning models.

5.8. Train-Test Split for Linear and Random Forest Models:

We are splitting the dataset into train/validation and test sets based on the month and year of the pickup. Specifically, all trips from October, November, and December 2024 are reserved for testing, while the rest are used for training/validation. We do this separately for the two types of preprocessed data: df_linear_encoded (OneHotEncoded for linear models) and df_tree (LabelEncoded for tree-based models). For each, X_* contains the features (all columns except the target total_amount) and y_* contains the target values. Finally, we print the shapes to confirm the number of samples and features in each split.

The split is time-based, with all trips except Oct–Dec 2024 (~98.9%) for training and Oct–Dec 2024 (~1.1%) for testing, ensuring the model is evaluated on future data. This is not a 70:30 random split; for that, train_test_split would be needed.

5.9. Baseline Model:

We build a baseline model by predicting each trip's fare as the historical average for similar trips, grouping by taxi_color, pickup_borough, dropoff_borough, month, day_of_week, and hour. It first computes these averages from the full dataset, then merges them with the train/validation set and calculates the RMSE to evaluate baseline performance. The same process is repeated for the test set (Oct–Dec 2024) to see how well this simple approach generalizes to future trips, providing a benchmark before applying machine learning models.

5.10. Linear Regression:

We train and evaluate a Linear Regression model to predict total_amount. First, it drops the pickup_datetime column and ensures all features are numeric, filling any non-numeric or missing values with zeros. The target variable is also cast to float. Then, the model is trained on the train/validation set (X_linear_train, y_linear_train), and predictions are made for both the training and test sets. Finally, the RMSE (Root Mean Squared Error) is calculated for both sets to assess how well the linear model fits the historical data and generalizes to unseen future trips.

5.10. Random Forest:

This code trains and evaluates a Random Forest Regressor to predict total_amount. It first removes the pickup_datetime column and ensures all features are numeric, filling any missing values with zeros. The model is then trained on the training set (X_tree_train, y_tree_train) and used to predict both the training and test sets. Finally, the RMSE is computed for both sets to measure the model's accuracy on the historical training data and its ability to generalize to unseen test data.

Section 6: Evaluation

6.1. Baseline Model Results:

The baseline model, which predicts the average fare per group, gives an RMSE of 9.46 on the train/validation set and 16.25 on the test set (Oct–Dec 2024). This shows that while it fits historical data moderately well, it performs worse on future trips, indicating that temporal variability affects fares and the simple baseline doesn't generalize. Any predictive model should aim to reduce the test RMSE below 16.25 to meaningfully improve over this baseline.

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Baseline RMSE (train/val): 9.4640
Baseline RMSE (test Oct-Nov-Dec 2024): 16.2501
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6.2. Linear Regression:

The Linear Regression model achieves an RMSE of 15.17 on the train/validation set and 25.69 on the test set, which is notably higher than the baseline RMSE (9.46 train, 16.25 test). This indicates that the linear model is overfitting less on the training data but struggles to generalize to the future period, likely because it cannot capture the nonlinear patterns and complex interactions present in taxi fare data. Overall, it underperforms compared to the baseline, suggesting that a more flexible model may be needed.

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Linear Regression RMSE (train/val): 15.1672
Linear Regression RMSE (test): 25.6887
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6.3. Random Forest:

The Random Forest model performs very well on the train/validation set with an RMSE of 2.30, showing it fits the training data closely. On the test set, the RMSE rises to 15.25, which is much closer to the baseline RMSE of 16.25, indicating good generalization to unseen data. Overall, the Random Forest captures complex patterns and nonlinearities effectively, outperforming the Linear Regression model and providing a robust improvement over the baseline.

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Random Forest RMSE (train/val): 2.3046
Random Forest RMSE (test): 15.2490
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6.3. Best Model:

The **Random Forest** is the best model among the three. It achieves the lowest RMSE on both train/validation (2.30) and test (15.25) sets, outperforming both the baseline and Linear Regression, which struggle to capture the nonlinearities in taxi fare data. While it is more computationally intensive than Linear Regression, the improvement in accuracy especially on

unseen test data justifies the added complexity. It also generalizes well to future trips, unlike Linear Regression, and offers a significant reduction in prediction error compared to the baseline, making it the most reliable choice for fare prediction

Section 7: Conclusion

This project offered a comprehensive exploration of the New York City taxi dataset, highlighting the full data engineering lifecycle: ingestion, cleaning, analysis, and predictive modeling. Starting with nearly a billion records across yellow and green taxis, we successfully standardized and merged datasets, removed unrealistic and duplicate records, and enriched the data with location and temporal features.

The analytical portion provided actionable business insights: peak travel times occur in the evenings and on weekends medium-duration trips (10–60 mins) offer the best balance of revenue and efficiency and Manhattan remains the highest revenue-generating borough, emphasizing the importance of targeted fleet allocation. Insights on tipping behaviour suggest that while tips are common, extremely high tips are rare, indicating opportunities for incentive programs or targeted driver strategies.

The machine learning analysis demonstrated the limitations of simpler models. While the baseline and Linear Regression models struggled to generalize to future trips, the Random Forest model captured nonlinear relationships and complex patterns in trip pricing, providing robust predictions with an RMSE of 15.25 on unseen data. This underscores the importance of model flexibility and feature engineering when dealing with large-scale, real-world data.

Overall, this assignment reinforced the value of careful data preparation, exploratory analysis, and model selection to derive meaningful insights and accurate predictions in urban mobility contexts.

Challenges Faced

1. Memory Constraints: Working with nearly a billion rows in Databricks Free Edition repeatedly caused crashes during sampling, joins, and aggregation. Multiple attempts at 5–10% sampling failed, forcing us to reduce the dataset to a 1% sample (~500,000 rows) for machine learning. This highlighted the importance of careful resource management when working with big data.

2. Data Standardization Across Sources: Yellow and green taxi datasets had differing column names, missing columns, and inconsistent formats. Merging them required careful renaming, creating placeholder columns, and joining with location reference data. Misalignment at this stage could have led to incorrect aggregations or feature misinterpretation.

3. Handling Outliers & Missing Values: Significant portions of the dataset contained extreme values or missing entries, particularly in numeric fields like trip_distance,

total_amount, and tip_amount. Decisions around capping outliers, imputing values, or leaving them untouched required balancing model accuracy against data representativeness.

4. Time-based Splitting for Predictive Modeling: Unlike random splits, splitting by future months (Oct–Dec 2024) to simulate realistic forecasting introduced additional complexity in validation. It demanded careful management of training/test sets, ensuring no data leakage while maintaining sufficient sample sizes for model learning.

Key Learnings

1. Data Cleaning at Scale: Handling nearly a billion records required structured cleaning pipelines, with separate validation after each step to ensure data integrity. Even small errors can propagate and significantly affect outcomes.

2. Feature Engineering Matters: Temporal features (hour, day, month) and location enrichments were critical in explaining fare variability and improving model performance.

3. Model Selection & Flexibility: Linear models may fail to capture complex, nonlinear relationships in large-scale real-world datasets. Tree-based ensemble models like Random Forests are more effective for such predictive tasks.

4. Business Insights Depend on Granular Analysis: Breaking down trips by duration, borough, and time revealed patterns that aggregate metrics would miss, emphasizing the value of detailed, structured aggregation for decision-making.